Name: _____

TOPIC TEST 2

Chapter 9: Derivatives

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 6 short answer questions (20 marks)
- Part C: 2 analysis questions (12 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 $\lim\limits_{x \to 10} \dfrac{2x - 20}{x^2 - 100} =$

A 0

B $\dfrac{1}{10}$

C $\dfrac{1}{2}$

D 2

E none of the above

[1 mark]

2 The gradient of the curve with equation $y = f(x)$ at the point $(a, f(a))$ is given by:

A $\lim\limits_{h \to 0} \dfrac{f(a+h) - a}{h}$

B $\dfrac{f(a+h) - f(a)}{h}$

C $\lim\limits_{h \to 0} \left[\dfrac{f(a+h)}{h} - f(a) \right]$

D $\lim\limits_{h \to 0} \dfrac{f(a+h) - f(h)}{h}$

E $\lim\limits_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$

[1 mark]

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3 If $y = \frac{3x^{10}}{5}$ then $\frac{dy}{dx}$ is equal to:

A $\frac{3x^9}{5}$

B $30x^9$

C $\frac{x^9}{6}$

D $6x^9$

E $\frac{3x^9}{50}$

[1 mark]

4 The equation of the tangent to $y = ax^2$ at the point $(2a, 4a^3)$ is

A $y = 2ax + 4a^3$

B $y = 2ax - a^2$

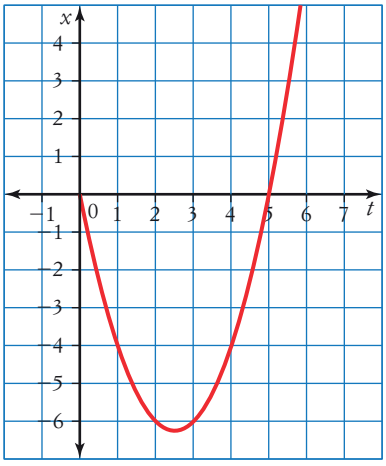
C $y = 3ax - 8a^2$

D $y = 2a^2x + 4a^2$

E $y = 4a^2x - 4a^3$

[1 mark]

5 The displacement–time graph of an object moving in a straight line over 4 minutes is shown below.



The velocity of the object is best described as:

A always positive

B initially negative, then changing to positive, then changing back to negative

C initially negative, then changing to and remaining positive

D initially positive, then changing to negative, then changing back to positive

E initially positive, then changing to and remaining negative.

[1 mark]

Part B

6 short answer questions
20 marks
Show your working where appropriate.

1 For the function $f(x) = 5x^2 - 8x + 1$:

a find $f(1)$

b find $f(1 + h)$

c find the gradient of the secant between the point where $x = 1$ and $x = 1 + h$

d hence, choose appropriate values for h to find $f'(1)$.

[4 marks]

2 Using first principles, find the derivative of $f(x) = -2x^2$

[3 marks]

3 By first considering the chord joining the points $P(x, y)$ and $Q(x + h, y + k)$, find the gradient of the curve $y = 2x^3$ at $x = 1$.

[4 marks]

4 A particle moves so that its distance (in cm) from a fixed point is given by $x = \frac{1}{7}t^7$, where t is the time in seconds. Find its velocity after 2s.

[2 marks]

5 Find the equation of the tangent to the curve $f(x) = 2x^3$ at the point (2, 16).

[4 marks]

6 A candle burns with a steady flame and gradually diminishes. The height of the candle in cm after burning for t seconds is given by the rule $x = -\frac{1}{4\pi}t^2$

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Find the exact speed at which the height of the candle is decreasing after it has been alight for 3 seconds.

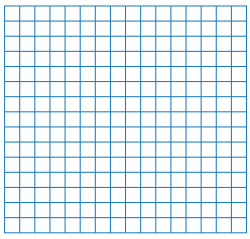
[3 marks]

Part C

2 analysis questions
12 marks
Show your working where appropriate.

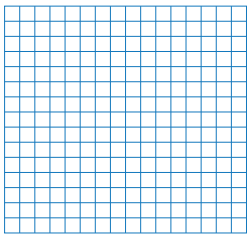
1 The position of a pebble thrown into the Yarra river is given by $x = -7t^2$, where x is in metres and t is in seconds.

a Draw a graph showing the height from 0 secs to 2 secs in 0.2 intervals.



b Calculate the speed, $\frac{dx}{dt}$, of the pebble.

c Draw a graph of the speed of the pebble from 0 secs to 2 secs.



d Calculate the average speed from 0 secs to 2 secs.

e Compare the average speed to the instantaneous speed.

[6 marks]

2 The cost of producing 60 ovens is given by:
$C(x) = 0.04x^3$, where the cost, $C(x)$, of producing x ovens is given in dollars.

a Find the cost of producing 60 ovens.

b Find the average cost of producing each oven if 60 ovens are made.

c Find the marginal cost of producing an extra oven over 60 ovens.

d Compare the average and marginal costs at:

i 60 ovens

ii 100 ovens

iii 130 ovens.

[6 marks]
Total marks: 37

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Answers

Part A

[1 mark each]

1 $\lim\limits_{x\to 10}\dfrac{2x-20}{x^2-100}=\lim\limits_{x\to 10}\dfrac{2(x-10)}{(x-10)(x+10)}=\lim\limits_{x\to 10}\dfrac{2}{x+10}=\dfrac{2}{20}=\dfrac{1}{10}$

$\therefore$ **B** **[1 mark]**

2 Replace x in $f'(x)=\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$ with a.

$\therefore$ **E** **[1 mark]**

3 Using the rule for finding the derivative of a polynomial:

$$y=\frac{3x^{10}}{5}\Rightarrow\frac{dy}{dx}=\frac{3\times 10\times x^{10-1}}{5}=6x^9$$

$\therefore$ **D** **[1 mark]**

4 The derivative of the function is $y'=2ax$. At the point $(2a, 4a^3)$, the tangent has gradient $4a^2$. The equation of the tangent in point slope form is:

$$y-4a^3=4a^2(x-2a)$$
$$y=4a^2x-8a^3+4a^3$$
$$=4a^2x-4a^3$$

$\therefore$ **E** **[1 mark]**

5 From 0 to about 2.5 seconds, the gradient of the graph is negative. The gradient is 0 where the graph turns, and after that it becomes positive.

$\therefore$ **C** **[1 mark]**

Part B

1 a $f(1)=5(1)^2-8(1)+1=-2$

[1 mark]

b $f(1+h)=5(1+h)^2-8(1+h)+1$
$=5(1+2h+h^2)-8-8h+1$
$=5+10h+5h^2-8-8h+1$
$=5h^2+2h-2$ **[1 mark]**

c $\dfrac{f(1+h)-f(1)}{h}=\dfrac{5h^2+2h-2-2}{h}$
$=5h+2$ **[1 mark]**

d $f'(1)=\lim\limits_{h\to 0}(5h+2)$
$=2$ **[1 mark]**

2 $f'(x) = \lim_{h \to 0}\left[\dfrac{-2(x+h)^2 - (-2x^2)}{h}\right]$ **[1 mark]**

$= \lim_{h \to 0}\left[\dfrac{-2x^2 - 4xh - 2h^2 + 2x^2}{h}\right]$ **[1 mark]**

$= \lim_{h \to 0}\left[-4x - 2h\right]$

$= -4x$ **[1 mark]**

3 Slope of chord $= \left[\dfrac{2(x+h)^3 - 2(x)^3}{h}\right]$

$= \left[\dfrac{6x^2h + 6xh^2 + 2h^3}{h}\right]$ **[1 mark]**

$= 6x^2 + 6xh + 2h$ **[1 mark]**

$f'(x) = \lim_{h \to 0}\left(6x^2 + 6xh + 2h^2\right)$

$= 6x^2$ **[1 mark]**

$f'(1) = 6$ **[1 mark]**

4 $v = t^6$ **[1 mark]**

For $t = 2$, $v = (2)^6 = 64$ cm/s **[1 mark]**

5 $f'(x) = 6(x)^2$ **[1 mark]**

$f'(2) = 6(2)^2 = 24$ **[1 mark]**

Using (2, 16) and $m = 24$ in $y = mx + c$

$16 = 24 \times 2 + c$ **[1 mark]**

So $c = -32$.

$y = 24x - 32$ **[1 mark]**

6 $v(t) = x'(t) = -\dfrac{1}{2\pi}t$ **[1 mark]**

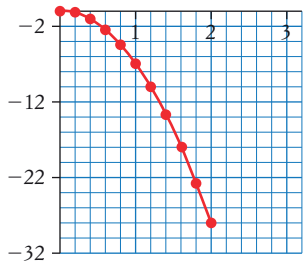
$v(3) = -\dfrac{1}{2\pi} \times 3 = -\dfrac{3}{2\pi}$ cm/s **[1 mark]**

The height is decreasing at a speed of $\dfrac{3}{2\pi}$ cm/s. **[1 mark]**

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Part C

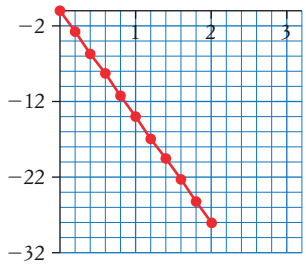
1 a



[1 mark]

b $\frac{dx}{dt} = -14t$ **[1 mark]**

c



[1 mark]

d Average speed $= \frac{\left[-7(2)^2\right] - \left[-7(0)^2\right]}{2 - 0}$ **[1 mark]**

$= \frac{-28}{2}$ **[1 mark]**

$= -14$ m/s **[1 mark]**

e The average speed is the average of the instantaneous speeds. **[1 mark]**

2 a $C(60) = 0.04(60)^3 = \$8640$ **[1 mark]**

b Average cost $= \frac{0.04(60)^3 - 0}{60 - 0} = \$144$ **[1 mark]**

c $C'(x) = 0.12(60)^2 = \$432$ **[1 mark]**

d i The average cost ($144) is lower than the marginal cost ($432).

ii The average cost ($400) is lower than the marginal cost ($1200).

iii The average cost ($676) is lower than the marginal cost ($2028).

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